A Dual Cross-Attention-Based Dynamic Weighting Fusion Method for Cross-Pair Landsat and MODIS Images

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***Abstract*—Spatiotemporal fusion technology plays a critical role in Earth observation by generating remote sensing imagery with both high spatial and temporal resolutions. However, most existing fusion models rely on three or more input images, which limits their practical applicability due to common challenges such as cloud contamination and data missing. To address these limitations, this study develops a dual cross-attention-based dynamic weighting fusion (DCAF) method that utilizes cross-pair Landsat and MODIS imagery. The proposed approach leverages two cross-attention mechanisms to facilitate mutual feature calibration across multisource and multiscene inputs. By explicitly modeling the intrinsic relationship between low- and high-resolution images, DCAF integrates adaptive weight allocation and hierarchical feature fusion strategies to effectively exploit the heterogeneous characteristics of multisource data. Experimental results demonstrate that, compared with four state-of-the-art fusion methods, DCAF delivers superior performance under a two-input setting and achieves accuracy comparable to that of three-input models. Furthermore, DCAF demonstrates strong generalization capability across heterogeneous landscapes, providing a practical and robust solution for enhancing the temporal and spatial resolution of remote sensing images.**

***Index Terms*—Cross-attention mechanism, dynamic weighting fusion, Landsat, MODIS, spatiotemporal fusion.**

I. Introduction

REMOTE sensing technology serves as an essential tool for large-scale Earth observation [1], [2], [3]. However, due to hardware limitations, satellite imagery exhibits an inherent tradeoff between spatial and temporal resolution [4], [5]. An increasing number of practical applications require satellite remote sensing data with both high spatial and temporal resolutions, such as phenology monitoring [6], [7], [8], land-cover change detection [9], [10], [11], water body dynamics analysis [12], [13], [14], and urbanization assessment [15], [16]. In such contexts, imagery with compromised spatial or temporal resolution often proves insufficient for accurate analysis [17], [18]. To address this limitation, spatiotemporal fusion techniques have emerged as a promising solution [14], [19], [20], [21]. These methods integrate multisource remote sensing data with complementary spatial and temporal characteristics to generate synthetic images with enhanced resolution in both dimensions [22], [23], thereby improving the applicability of remote sensing in dynamic monitoring tasks.

Current spatiotemporal fusion methods can be broadly classified into four categories [24]: 1) weight function-based methods; 2) unmixing-based methods; 3) hybrid methods; and 4) learning-based methods.

Weight function-based methods estimate high-resolution pixel values by computing a weighted combination of all available input images. The most representative model is the spatial and temporal adaptive reflectance fusion model (STARFM) [20], which combines spectral variation information of similar pixels through a weighting function to generate high-resolution fusion data. To improve performance in heterogeneous landscapes, the enhanced STARFM [25] introduces adaptive weighting by assigning distinct transformation coefficients to homogeneous and heterogeneous regions. The image inpainting and steering kernel regression fusion model [26] refines the weights of neighboring pixels and enhances the detection of land-cover changes. Weight function-based methods are characterized by their simple parameterization and high computational efficiency. However, they often struggle to capture abrupt surface changes or dynamic landform transformations, leading to reduced prediction accuracy in heterogeneous environments.

Unmixing-based methods are grounded in the linear spectral mixing theory, which assumes that the reflectance of a coarse-resolution pixel can be expressed as a linear combination of spectral responses from various land-cover classes. These methods aim to decompose coarse-resolution pixels into their constituent components and estimate corresponding fine-resolution pixel values [27], [28], [29]. One of the earliest models in this category, the multisensor multiresolution technique [30], suffers from considerable uncertainty in the decomposition of mixed pixels and often fails to accurately capture intraclass variability [31]. To overcome these challenges, improved methods have been developed. For example, the spatial–temporal data fusion approach [32] integrates both

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spatial variation and nonlinear temporal variation to unmix the temporal dynamics of coarse-resolution pixels for fine-resolution prediction. Although unmixing-based methods offer a solid theoretical foundation, they typically rely on prior land-cover classification, and classification uncertainties can propagate into the fusion process, degrading overall accuracy.

Hybrid methods combine the advantages of weight function-based and unmixing-based methods to enhance fusion performance. A representative example is the flexible spatiotemporal data fusion (FSDAF) algorithm [33], which combines principles from both paradigms to enable more flexible and accurate mapping across space and time. Similarly, Li et al. [34] proposed a spatiotemporal fusion model for remotely sensed images and land-cover maps that integrates spectral unmixing with Bayesian inference to produce high-resolution land-cover maps.

Learning-based methods have recently emerged as a powerful class of spatiotemporal fusion techniques capable of modeling complex nonlinear relationships between multisensor data to generate high-resolution target images [35], [36]. Among these, sparse representation learning methods employ paired dictionaries to learn mapping relationships between fine- and coarse-resolution images at reference dates, which are then used to infer high-resolution data at target dates [37], [38], [39], [40]. Although effective in capturing land-cover changes, these methods may introduce geometric distortions in ground objects [41]. Bayesian-based approaches formulate spatiotemporal fusion as a maximum a posteriori estimation problem, explicitly modeling the probability distribution of the target data. Specifically, the spatiotemporal Bayesian data fusion method [42] establishes mathematical relationships among input images to better capture temporal dynamics, while the highly scalable temporal adaptive reflectance fusion model [43] extends this framework by incorporating bias-aware Kalman filtering to generate high-precision reflectance data with reduced noise. Despite their flexibility, these methods remain sensitive to input data quality. Deep learning approaches have demonstrated strong potential in spatiotemporal fusion, owing to their powerful nonlinear fitting and representation learning capabilities [44], [45]. For example, Liu et al. [46] incorporated a temporal constraint mechanism into a two-stream convolutional neural network to predict target images, while Zhang et al. [31] proposed a generative adversarial network-based spatiotemporal fusion framework that improves prediction accuracy through an end-to-end two-stage process. More recently, Lin et al. [47] developed the cross-attention adaptive weighting fusion network (CAFE), which employs a cross-attention mechanism to enhance fusion accuracy, particularly in capturing dynamic changes in ground objects.

However, most existing spatiotemporal fusion models require at least three input images and are primarily developed and evaluated using idealized datasets. These approaches often overlook practical challenges such as regional data gaps due to cloud contamination, missing observations, and sensor limitations. To mitigate these issues, recent studies have increasingly focused on dual-input spatiotemporal fusion networks [45], [48]. For example, Tan et al. [49] enhanced spatiotemporal fusion performance by incorporating conditional generative adversarial networks with advanced normalization strategies. Furthermore, the enhanced cross-paired wavelet-based spatiotemporal fusion network (ECPW-STFN) [50] integrated wavelet transforms with a generative adversarial network to better preserve both global and local features, thereby substantially improving the accuracy of dual-input spatiotemporal fusion. Nevertheless, there remains considerable potential for enhancing the performance of dual-input fusion approaches.

While existing three-input fusion networks have demonstrated strong performance, two-input fusion scenarios still offer considerable room for improvement, particularly when high-quality optical imagery at multiple time points is difficult to obtain. To address this challenge, this study focuses on Landsat–MODIS cross-pairs and proposes a dual-input, dual-cross-attention-based dynamic weighting fusion (DCAF) method to effectively integrate multiscale remote sensing information. To fully exploit the complementary information inherent in multisource data, the proposed approach introduces an enhanced dual-layer cross-attention mechanism that enables mutual calibration between feature maps. Moreover, a residual dense attention module (RDAM) is employed to extract multilevel features, effectively capturing intrinsic relationships between low- and high-resolution images. Finally, multilevel adaptive feature fusion is applied to predict high-spatiotemporal-resolution images, yielding superior fusion accuracy and robust performance over diverse and complex surface conditions.

II. Methodology

The overall framework of the proposed spatiotemporal fusion method is illustrated in Fig. 1. Compared with the cross-attention-based fusion framework of Lin et al. [47], we develop an enhanced model tailored for practical application, where acquiring multitemporal cloud-free optical imagery is often challenging. Specifically, the original three-input network is reformulated into a two-input fusion scheme that relies solely on a fine-resolution reference image and a coarse-resolution target image. This cross-paired input configuration captures inherent temporal asymmetry and motivates the design of modules that compensate for the lack of temporal guidance.

To fully leverage multiscale information and capture high-level heterogeneous features, this study proposes the DCAF model to estimate high-resolution imagery at the target date. The proposed framework integrates two cross-attention modules—multichannel deep cross-attention (MDCA) and global feature cross-attention (GFCA)—to extract and recalibrate feature maps. An RDAM is further employed to enhance feature representation, while a difference weighting fusion module (DWFM) is designed to effectively model land-cover dynamics over time. All modules are specifically designed to operate effectively under the two-input scenario. Detailed descriptions of each module are provided in Sections II-A–II-C.

A. Double Cross-Attention Mechanism

High-resolution images provide fine spatial detail, whereas coarse-resolution images offer superior spectral fidelity [51].

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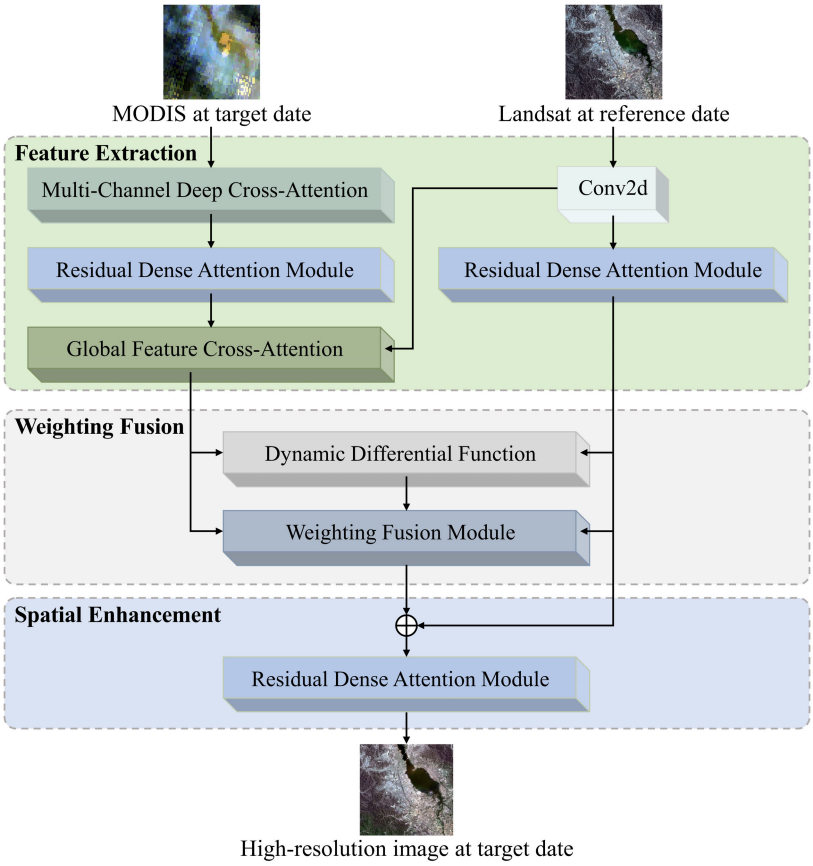


Fig. 1. Overall architecture of the proposed DCAF model.

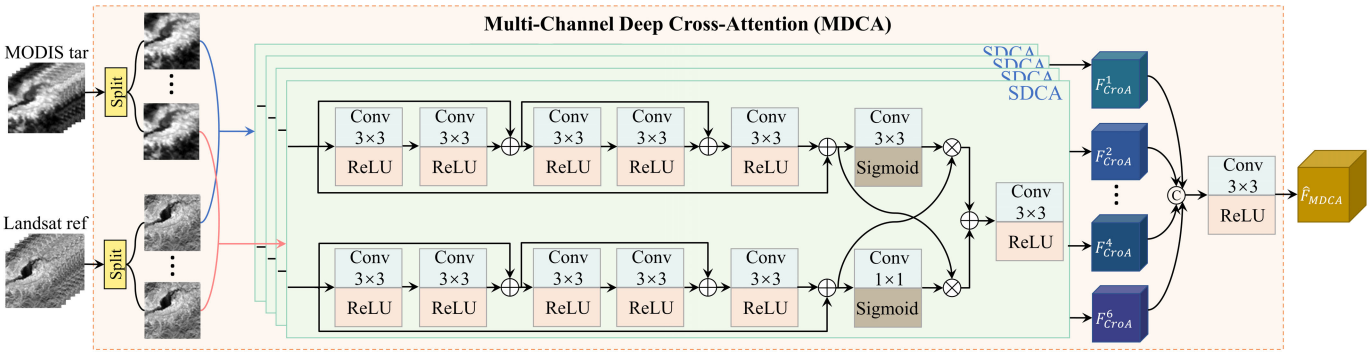


Fig. 2. Architecture of the multichannel deep cross-attention (MDCA) module.

To effectively integrate complementary information from only two input images, the proposed framework employs a dual cross-attention mechanism [52], [53] that dynamically recalibrates multiscale feature representations, specifically designed for scenarios with limited temporal guidance. Two key modules are introduced: MDCA and GFCA. In particular, MDCA performs band-wise spatial and spectral weighting calibration, whereas GFCA captures intrinsic interband dependencies to facilitate more effective and coherent feature integration.

As illustrated in Fig. 2, the cross-attention mechanism includes a channel attention block for low-resolution MODIS images (LMCA) and a spatial attention block for high-resolution Landsat images (HLSA). Unlike conventional cross-attention mechanisms and the multichannel separated cross-attention (MSCA) in Lin et al. [47], the proposed MDCA incorporates a single-band deep cross-attention (SDCA) block with hierarchical convolutions and skip connections, enabling the extraction of deeper, context-rich features under the constraint of a two-input setting. As depicted in Fig. 2, the SDCA block adopts a hierarchical convolutional structure with skip connections, allowing for progressive feature aggregation while mitigating the vanishing gradient problem and preserving the integrity of the original input features. Overall, this design facilitates coarse-to-fine cross-modal alignment between MODIS and Landsat features. Meanwhile, the introduced skip connections help preserve low-level

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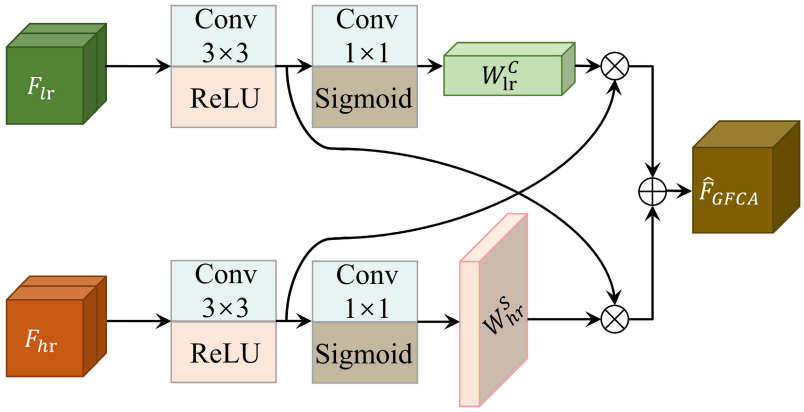


Fig. 3. Architecture of the GFCA module.

information and stabilize cross-temporal feature calibration, thereby improving both spatial detail reconstruction and spectral consistency.

In the LMCA block, low-resolution MODIS features are normalized using a sigmoid activation function to generate channel weights, which are then applied to modulate the high-resolution Landsat features. Simultaneously, the HLSA block extracts spatial weights from the high-resolution Landsat features to recalibrate the low-resolution features. The recalibrated features are then fused through element-wise addition, effectively integrating the magnitude variations captured by the coarse-resolution data with the spatial details provided by the fine-resolution data. This synergy enhances feature representation even under reduced temporal input. The deep cross-attention mechanism can be expressed as

$$F_{HLSA} = \left(W_{HLSA1} \circ Conv_{lr}^{skip}(Flr) + b_{HLSA1}\right) \otimes \sigma(W_{HLSA2} \circ Conv_{hr}^{skip}(Fhr) + b_{HLSA2}) \quad (1)$$

$$F_{LMCA} = \left(W_{LMCA1} \circ Conv_{hr}^{skip}(Fhr) + b_{LMCA1}\right) \otimes \sigma(W_{LMCA2} \circ Conv_{lr}^{skip}(Flr) + b_{LMCA2}) \quad (2)$$

$$F_{CroA} = f_{cat}(F_{LMCA}, F_{HLSA}) \quad (3)$$

where F_{HLSA}, F_{LMCA}, and F_{CroA} denote the output feature maps of the HLSA block, the LMCA block, and the cross-attention module, respectively. $Conv_{lr}^{skip}(Flr)$ and $Conv_{hr}^{skip}(Fhr)$ represent convolution operations with skip connections applied to the low-resolution and high-resolution feature maps, respectively. The symbols $\circ$ and $\otimes$ denote convolution and element-wise multiplication operations, respectively. f_{cat} represents a concatenation operation. σ is the sigmoid function. W is the weight parameter. b is the bias term of each attention block.

The strong interband correlations inherent in multispectral data make it challenging to fully characterize complex cross-channel relationships through independent per-band processing. To address this challenge in the two-input scenario, the GFCA module [47] is introduced, as illustrated in Fig. 3. This module incorporates cross-attention to dynamically model intrinsic interband dependencies and ensure coherent multiband feature representation despite limited temporal guidance.

MDCA primarily focuses on band-wise spatial–spectral interactions between MODIS and Landsat features. Through the SDCA block, which integrates hierarchical convolutions and skip connections, cross-attention is applied independently to each spectral band. This design enables fine-grained alignment while enhancing local spatial detail reconstruction and band-level feature recalibration. In contrast, the GFCA module operates on aggregated features after MDCA, performing cross-attention across all feature channels. This mechanism captures intrinsic interband dependencies and high-level spectral correlations, thereby ensuring coherent multiband feature representation and improving global spectral consistency. The combination of MDCA and GFCA allows the network to simultaneously capture local spatial details and global spectral consistency within a two-input framework, leading to improved high-resolution reconstruction quality.

Unlike the attention mechanism in Lin et al. [47], the proposed MDCA employs an SDCA block with hierarchical convolutions and skip connections, enabling progressive feature aggregation and residual feature propagation within each band. In addition, GFCA integrates spatial–spectral cross-attention into a unified framework, more effectively capturing interband dependencies and enhancing the consistency of multiband feature representations.

B. Residual Dense Attention Module

Following feature map recalibration, the RDAM [52], [54] is employed to further optimize feature extraction under limited temporal guidance. As illustrated in Fig. 4, the RDAM integrates densely connected attention mechanisms to capture fine spatial details, while residual learning facilitates information propagation and improves training efficiency. The module comprises six residual channel–spatial attention (RCSA) blocks [55], in which local skip connections are utilized to transfer feature information and promote feature reuse.

Each RCSA block integrates dual-attention subblocks to recalibrate feature maps via channel and spatial weighting operations. The channel attention subblock employs global

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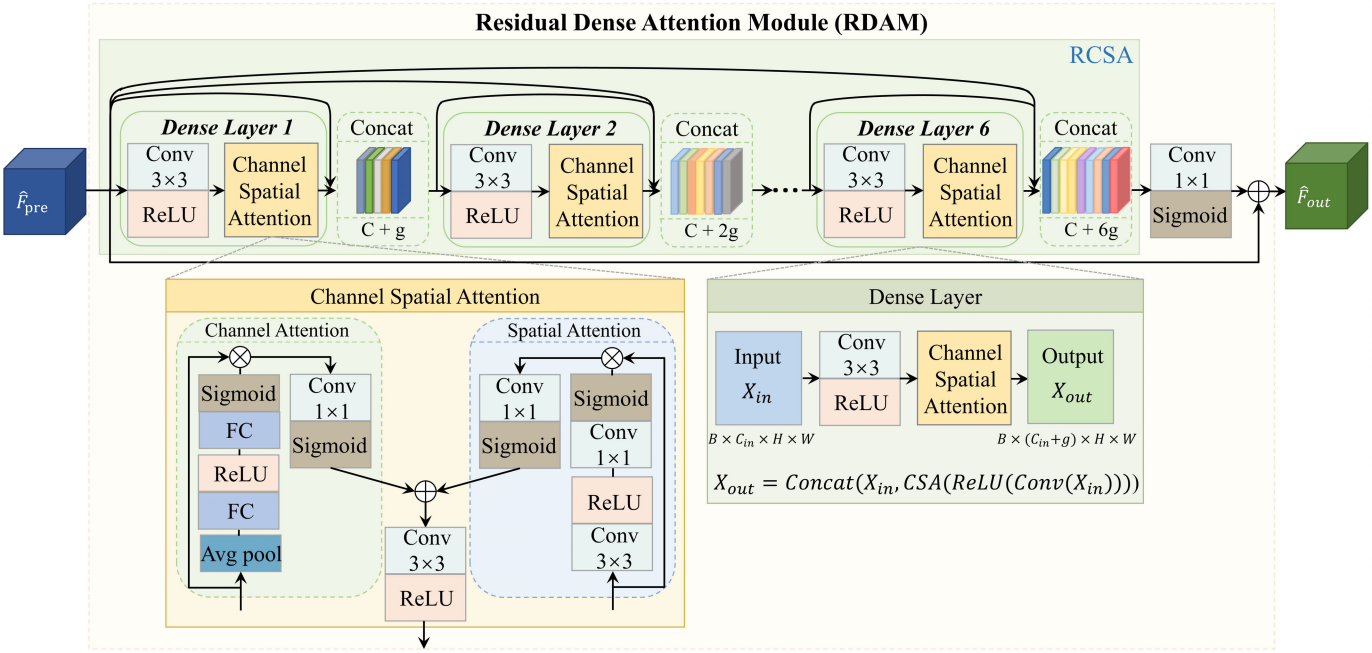


Fig. 4. Architecture of the RDAM.

average pooling to emphasize informative channels, while the spatial subblock utilizes depthwise convolution to enhance discriminative spatial regions. The outputs of these attention branches are fused via element-wise addition and further refined through local residual learning. Additionally, a global residual connection is incorporated to preserve the original input information. This comprehensive design compensates for reduced temporal redundancy in two-input fusion, substantially enhancing feature representation and network stability.

C. Temporal Difference Weighting Fusion

To adaptively model temporal variations using only two input images, a DWFM is designed. Instead of directly relying on predefined temporal differences, it captures local temporal dynamics by learning weighting coefficients from feature discrepancies between the recalibrated MODIS and Landsat feature maps. These weights are calculated through a sigmoid activation function, dynamically modulating feature contributions at each spatial location. Specifically, the DWFM follows a simple but effective rule: weights approaching 1 prioritize MODIS features, emphasizing areas with significant temporal variation, whereas weights approaching 0 retain more Landsat features, thereby preserving stable areas. This adaptive weighting mechanism effectively balances change detection and detail preservation, while enhancing robustness to noise and temporal inconsistencies in two-input scenarios

$$M = f_{\sigma}^{BN}\left(W \circ \left(\widehat{F}_{lr} - \widehat{F}_{hr}\right)\right) \tag{4}$$

$$F_{pre} = M \cdot \widehat{F}_{lr} + (1 - M) \cdot \widehat{F}_{hr} \tag{5}$$

where $\widehat{F}_{lr}$ and $\widehat{F}_{hr}$ denote the recalibrated MODIS and enhanced Landsat feature maps, respectively. W is the convolution kernel. $f_{\sigma}^{BN}(\cdot)$ represents the combined batch normalization and sigmoid function. M is the weight map. F_{pre} is the preliminary fusion result.

D. Loss Function

The overall loss function includes the Charbonnier loss function ($\mathcal{L}_{Charbonnier}$) [56], the edge loss function ($\mathcal{L}_{edge}$) [57], and the structural loss function ($\mathcal{L}_{structure}$). Joint optimization of these three loss functions simultaneously enhances pixel-level accuracy, edge preservation, and structural similarity. The total loss is defined as

$$\mathcal{L}_{total}(\Theta) = \alpha \cdot \mathcal{L}_{Charbonnier}(\Theta) + \beta \cdot \mathcal{L}_{edge}(\Theta) + \gamma \cdot \mathcal{L}_{structure}(\Theta) \tag{6}$$

where Θ denotes the network parameter. α, β, and γ are regularization parameters used to adjust the weights, which are adaptively determined by the following equations:

$$\alpha = \frac{\mathcal{L}_{Charbonnier}(\Theta)}{\mathcal{L}_{Charbonnier}(\Theta) + \mathcal{L}_{edge}(\Theta) + \mathcal{L}_{structure}(\Theta)} \tag{7}$$

$$\beta = \frac{\mathcal{L}_{edge}(\Theta)}{\mathcal{L}_{Charbonnier}(\Theta) + \mathcal{L}_{edge}(\Theta) + \mathcal{L}_{structure}(\Theta)} \tag{8}$$

$$\gamma = \frac{\mathcal{L}_{structure}(\Theta)}{\mathcal{L}_{Charbonnier}(\Theta) + \mathcal{L}_{edge}(\Theta) + \mathcal{L}_{structure}(\Theta)}. \tag{9}$$

Among them, the Charbonnier loss function ensures training stability while maintaining robustness to outliers. Its formulation is as follows:

$$\mathcal{L}_{Charbonnier}(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left\|(L_t^i - M_t^i) - \xi(M_t^i, L_r^i)\right\|^2 + \varepsilon^2} \tag{10}$$

where L_t denotes the label data, corresponding to the actual Landsat image at the target date, M_t represents the input

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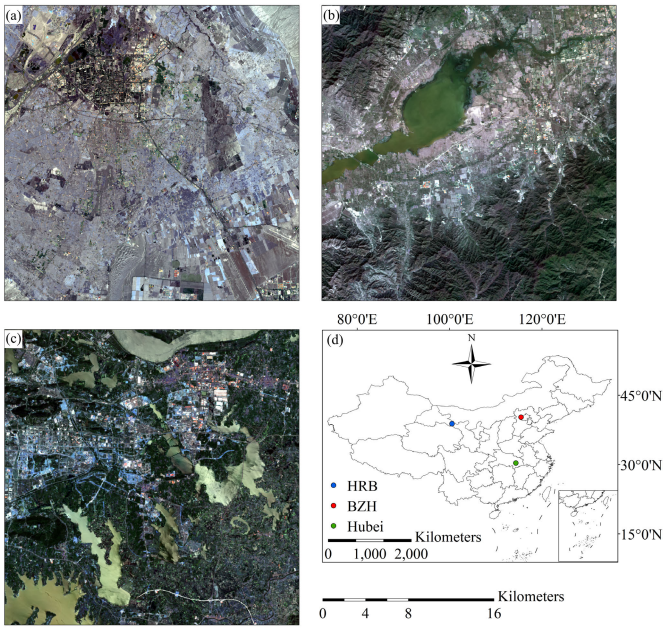


Fig. 5. Overview of the study areas. (a) RGB map of the Heihe River Basin (HRB), (b) RGB map of the Beijing-Zhangjiakou boundary area in Hebei Province (BZH), (c) RGB map of the area in Hubei Province, and (d) geographic locations of the three study areas.

MODIS image at the target date, and L_r refers to the input Landsat image at the reference date. N is the number of training image pairs. $\xi(\cdot)$ denotes the fusion network. ε is a constant term set to 0.001.

The edge loss function is used to effectively extract edge and textural information. The formula can be defined as

$$\mathcal{L}_{edge}(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left\|\Delta\left(L_t^i - M_t^i\right) - \Delta\left(\xi\left(M_t^i, L_r^i\right)\right)\right\|^2 + \varepsilon^2} \tag{11}$$

where $\Delta(\cdot)$ represents the Laplacian operator.

The structural loss function is based on the multiscale structural similarity [58] and is formulated as

$$\mathcal{L}_{structure}(\Theta) = 1 - \left[l_M\left(M_t^i, L_t^i\right)\right]^{\upsilon_M} \cdot \prod_{j=1}^{M}\left[c_j\left(M_t^i, L_t^i\right)\right]^{\rho_j}\left[s_j\left(M_t^i, L_t^i\right)\right]^{\mu_j} \tag{12}$$

where $l_M(\cdot)$ denotes the luminance consistency at the Mth scale; $c_j(\cdot)$ and $s_j(\cdot)$ represent the contrast similarity and structural similarity at the jth scale, respectively; and υ_M, ρ_j, and μ_j denote the corresponding weighting factors.

III. Experiments

A. Study Area and Datasets

In this study, three Landsat–MODIS paired datasets were constructed across geographically and environmentally distinct regions: 1) an area near the boundary between Beijing and Zhangjiakou in Hebei Province (BZH); 2) an area within the Heihe River Basin (HRB); and 3) an area in Hubei Province, as shown in Fig. 5. These regions encompass diverse land-cover types, including cropland, vegetation, bare soil, impervious surfaces, and water bodies, with increasing surface complexity from BZH to Hubei. Additionally, they contain both artificial water bodies (e.g., reservoirs) and natural water bodies (e.g., lakes and rivers), characterized by pronounced spatial heterogeneity and temporal variability. The diversity of land cover and the complexity of spatiotemporal dynamics make these datasets well-suited for evaluating the applicability and robustness of remote sensing spatiotemporal fusion methods under varying environmental conditions. In total, the BZH, HRB, and Hubei datasets comprise 48 Landsat–MODIS image pairs, with acquisition dates listed in Table I.

All three datasets share a consistent spatial coverage of 30×30 km^2. High-resolution imagery was acquired from the Landsat 8 Operational Land Imager (OLI), which offers a spatial resolution of 30 m and a revisit interval of 16 days. For coarse-resolution data, we employed MODIS surface reflectance products, including MOD09GA (500-m spatial resolution, daily revisit) and MOD09GQ (250-m spatial resolution, daily revisit). Before fusion, MODIS images were geometrically co-registered to the Landsat reference and spectrally aligned to ensure radiometric consistency. To improve the spatial fidelity of the fused outputs, the 250-m resolution red and near-infrared bands from MOD09GQ were used to replace their corresponding 500-m bands in MOD09GA, thereby preserving finer spatial details. Following band selection, all MODIS bands were uniformly resampled to 30-m resolution to ensure spatial consistency with the Landsat data before model input. This unified-resolution preprocessing mitigates potential spatial inconsistencies arising from mixed-resolution inputs and facilitates more reliable cross-sensor feature integration.

B. Experimental Design

To evaluate the effectiveness of the proposed method, we compared it with four widely used spatiotemporal fusion methods, including STARFM [20], FSDAF [33], CAFE [47], and ECPW-STFN [50]. Among these, only ECPW-STFN—similar to the proposed method—utilizes two input images for fusion, whereas the other models require at least three. All baseline methods were implemented using the default parameter settings specified in their original publications. In this study, the three regional datasets (BZH, HRB, and Hubei) were combined to construct a unified training dataset, rather than being used for independent region-specific training. Specifically, samples from all regions were jointly utilized during the training process, enabling the model to learn diverse spatiotemporal patterns across varying environmental conditions. During evaluation, model performance is reported separately for each region, allowing assessment of both region-specific adaptability and cross-regional generalization.

Fusion performance was quantitatively evaluated using four metrics: the correlation coefficient (CC), root-mean-square error (RMSE), mean absolute error (MAE), and structural similarity (SSIM) [59]. Among these indicators, CC and SSIM primarily measure the spectral fidelity and structural similarity of the images, with higher values indicating better fusion quality. In contrast, the RMSE and MAE are used to evaluate the absolute error in pixel values, where lower values denote higher accuracy.

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TABLE I
SUMMARY OF IMAGE INFORMATION FOR THE TRAINING AND TEST DATASETS

Dataset	Train		Test	
	Landsat (MM/DD/YY)	MODIS (MM/DD/YY)	Landsat (MM/DD/YY)	MODIS (MM/DD/YY)
BZH	09/25/2019	10/20/2019	11/28/2019	04/29/2020
	10/20/2019	11/12/2019	04/29/2020	10/22/2020
	11/12/2019	04/04/2020	10/22/2020	05/18/2021
	04/04/2020	10/13/2020	05/18/2021	04/26/2022
	10/13/2020	05/02/2021		
	05/02/2021	06/19/2021		
	06/19/2021	05/28/2022		
	05/28/2022	09/26/2022		
	09/26/2022	03/28/2023		
HRB	07/19/2018	11/24/2018	03/02/2020	03/21/2021
	11/24/2018	04/17/2019	03/21/2021	05/24/2021
	04/17/2019	05/03/2019	05/24/2021	11/03/2022
	05/03/2019	05/19/2019	11/03/2022	04/28/2023
	05/19/2019	07/22/2019	04/28/2023	09/03/2023
	07/22/2019	11/27/2019		
	11/27/2019	09/10/2020		
	09/10/2020	05/08/2021		
	05/08/2021	06/12/2022		
	06/12/2022	11/19/2022		
	11/19/2022	06/15/2023		
Hubei	10/30/2017	12/10/2017	04/08/2018	09/08/2018
	12/10/2017	02/12/2018	09/08/2018	10/26/2018
	02/12/2018	04/17/2018	10/26/2018	10/20/2019
	04/17/2018	09/15/2018	10/20/2019	11/14/2019
	09/15/2018	05/22/2019	11/14/2019	02/18/2020
	05/22/2019	10/29/2019		
	10/29/2019	12/07/2019		
	12/07/2019	04/13/2020		
	04/13/2020	08/03/2020		
	08/03/2020	10/22/2020		

This study implements the DCAF network using the PyTorch framework, with model training performed on an NVIDIA GeForce RTX 4070 GPU. The Adam optimizer was used to train the proposed model, with an initial learning rate set to 0.0001. There were 100 epochs and a batch size of 64 in the training process. The training data consisted of multitemporal MODIS target images and Landsat reference images, each comprising six spectral bands. A total of 32 640 training samples were generated through a patch-based processing strategy.

IV. RESULTS

A. Qualitative Analysis

The fusion results generated by different methods for the BZH area on April 26, 2022, are presented in Fig. 6. All methods generally captured the temporal changes; however, noticeable differences in reconstruction quality were observed. Specifically, STARFM suffered from spectral distortion due to limitations in its weighting strategy, whereas FSDAF alleviated this issue to some extent but still exhibited a noticeable speckle effect. The ECPW-STFN method exhibited spatial blurring, particularly in heterogeneous regions, reflecting its limited ability to preserve feature boundaries. In contrast, the fusion results from CAFE and the proposed DCAF method showed higher consistency with the reference image, benefiting from weighting functions that effectively incorporate land-cover change information from the input data. Furthermore, the zoomed-in subregions demonstrate that DCAF performed best in preserving both spatial textures and spectral fidelity. Overall, the visual evaluation indicates that the proposed DCAF method achieved the highest consistency with the ground truth, demonstrating superior spatial detail retention and spectral accuracy.

The fusion results from different methods for the HRB area on April 28, 2023, are shown in Fig. 7. The proposed DCAF method demonstrates the highest spatial and spectral consistency with the ground truth, particularly excelling in accurately capturing and preserving surface feature details. In contrast, although the other methods reflected changes in ground objects to varying degrees, they exhibited limitations in maintaining information integrity. Traditional approaches, STARFM and FSDAF, generated prominent red artifacts and introduced considerable spectral noise in fine-scale details. CAFE

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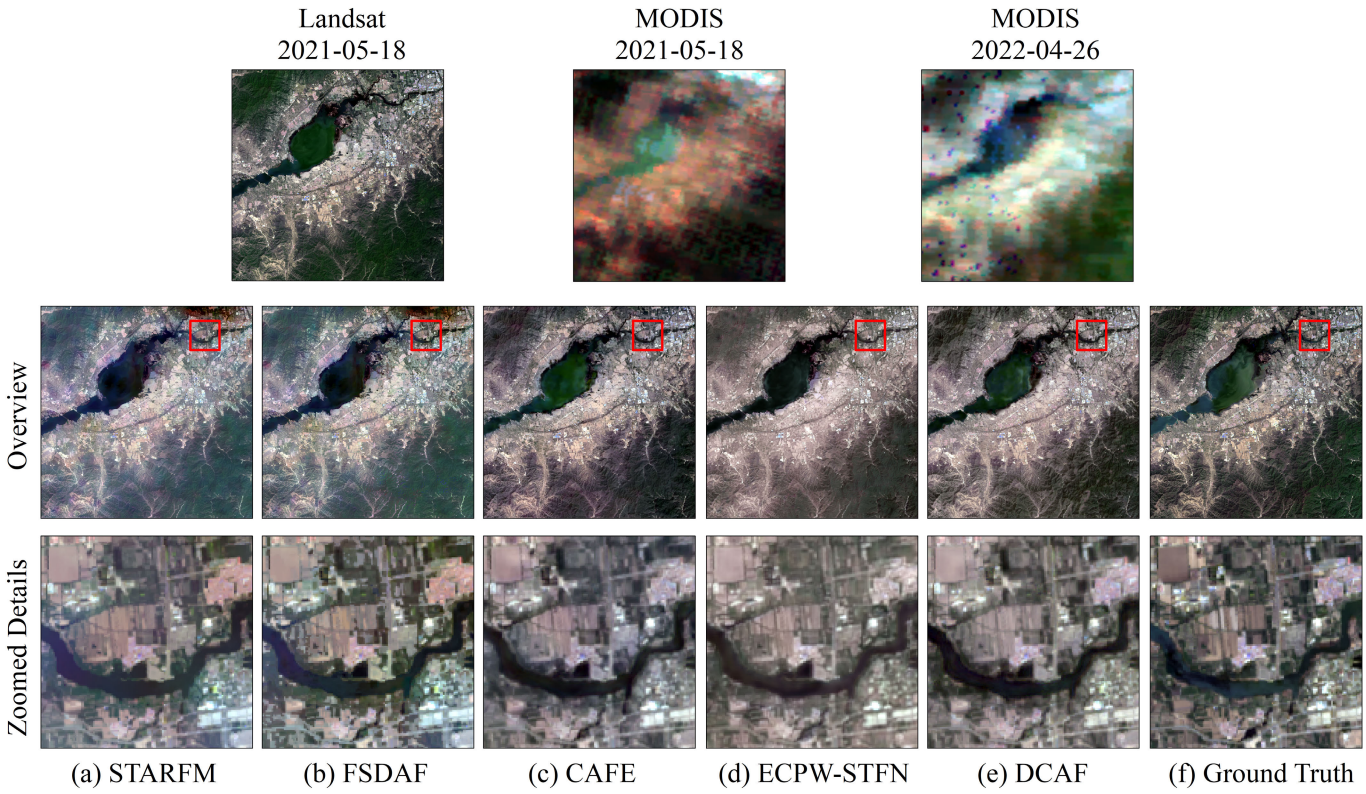


Fig. 6. Fusion results for the BZH area on April 26, 2022, obtained using different spatiotemporal fusion methods. The first row shows the input data, the second row presents the RGB composites of the fusion results, and the third row displays enlarged subregions highlighted by red boxes in the second row. (a) STARFM. (b) FSDAF. (c) CAFF. (d) ECPW-STFN. (e) DCAF. (f) Ground truth.

struggled to delineate object boundaries clearly, leading to blurred spatial structures. ECPW-STFN, affected by information loss, failed to adequately capture vegetation changes and to restore spectral details in built-up areas. Overall, the visual assessment confirms that DCAF outperformed the comparison methods in both spectral fidelity and texture preservation. These findings further highlight the adaptability and robustness of the DCAF method across diverse study areas.

The HRB is a typical oasis–desert transition region in arid northwestern China, characterized by highly heterogeneous land cover, including irrigated cropland, sparse vegetation, bare soil, built-up areas, and rivers. These land-cover types are distributed in fragmented patterns and exhibit strong seasonal dynamics driven by vegetation phenology and irrigation activities, resulting in pronounced spatial and temporal variability in surface reflectance. Under such conditions, the coarse spatial resolution of MODIS data leads to significant mixed-pixel effects, posing challenges for traditional fusion methods in accurately reconstructing fine-scale spatial structures and spectral variations. In contrast, the proposed DCAF method effectively integrates multiscale spatial–spectral features and adaptively captures contextual information, thereby improving reconstruction of spatial textures and spectral features, particularly in heterogeneous farmland–bare soil transition zones and built-up areas within the HRB.

The fusion results for the complex surface in the Hubei area on November 14, 2019, are displayed in Fig. 8. Overall, the results highlight significant limitations of traditional methods in this region. Specifically, STARFM lacks nonlinear modeling capability, resulting in pronounced speckle artifacts and poor performance in capturing heterogeneous landform changes. FSDAF, due to its parameter sensitivity, distorted spatial details and produced an over-sharpened fused image. In contrast, the deep-learning-based methods—CAFE, ECPW-STFN, and the proposed DCAF—exhibited stronger adaptability to complex landscapes, effectively capturing variations in landform features. The enlarged subregions further demonstrate that DCAF achieved the most accurate reconstruction of spatial details and clearly delineated boundaries between different landform types. While CAFE and ECPW-STFN captured the overall landform patterns, they exhibited reduced precision in spatial textures and boundary preservation, particularly in complex areas such as water-land interfaces and dense urban regions. These observations further confirm the superior performance of the DCAF method in handling complex surface conditions.

B. Quantitative Analysis

The quantitative evaluation results for the BZH, HRB, and Hubei datasets are summarized in Table II, which reports the average metric values of CC, RMSE, MAE, and SSIM. Traditional methods, STARFM and FSDAF, exhibited comparable performance but fell short of deep learning-based methods, highlighting the superior capability of deep

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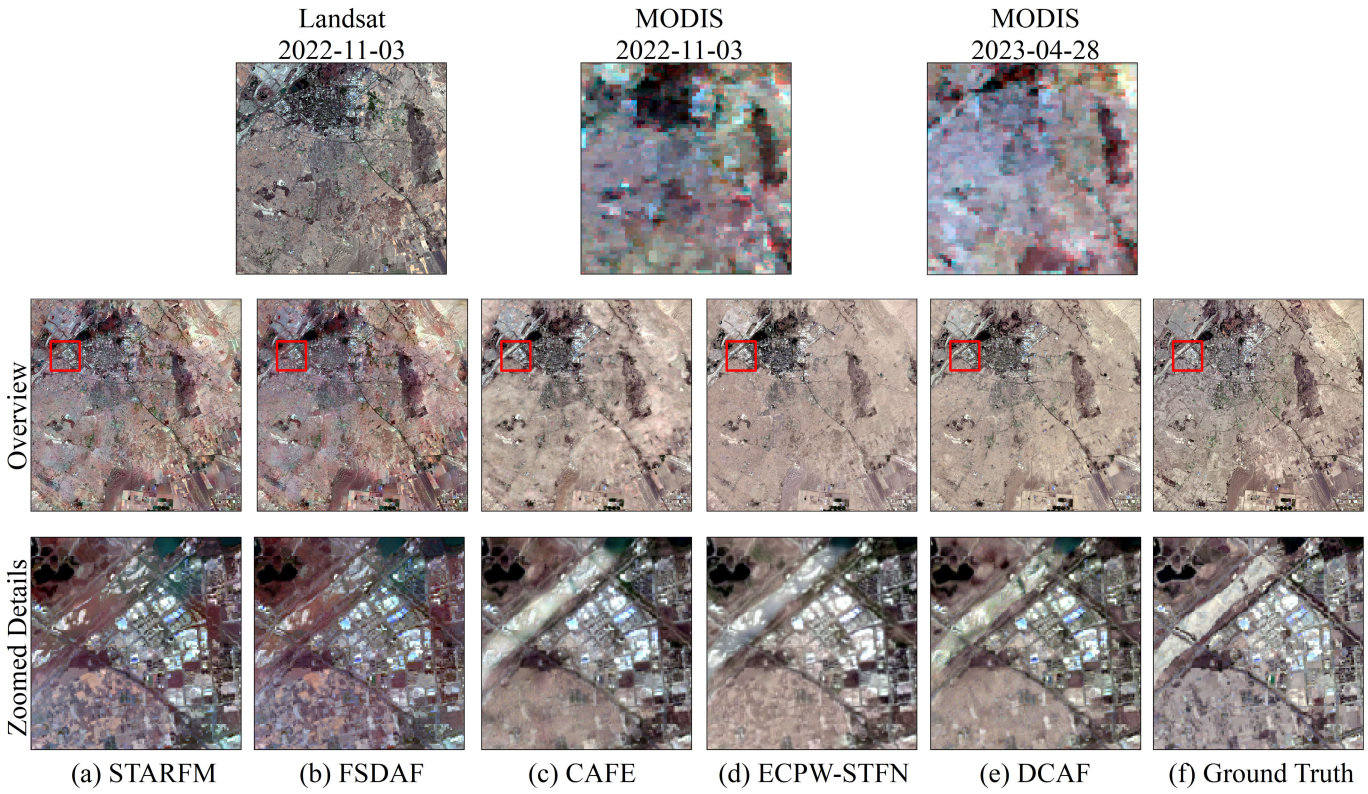


Fig. 7. Fusion results for the HRB area on April 28, 2023, obtained using different spatiotemporal fusion methods. The first row shows the input data, the second row presents the RGB composites of the fusion results, and the third row displays enlarged subregions highlighted by red boxes in the second row. (a) STARFM. (b) FSDAF. (c) CAFF. (d) ECPW-STFN. (e) DCAF. (f) Ground truth.

TABLE II
AVERAGE QUANTITATIVE EVALUATION RESULTS FOR THE THREE STUDY AREAS USING DIFFERENT MODELS

Area	Method	CC	RMSE	MAE	SSIM
BZH	STARFM	0.8423	0.0550	0.0484	0.7267
	FSDAF	0.8426	0.0549	0.0481	0.7232
	CAFE	**0.8953**	0.0322	0.0253	0.8058
	ECPW-STFN	0.8850	0.0309	0.0239	0.7981
	DCAF	0.8946	**0.0300**	**0.0231**	**0.8138**
HRB	STARFM	0.6724	0.0541	0.0448	0.6435
	FSDAF	0.6622	0.0549	0.0453	0.6191
	CAFE	0.7161	0.0329	0.0247	0.7166
	ECPW-STFN	0.7045	0.0347	0.0268	0.7045
	DCAF	**0.7400**	**0.0317**	**0.0234**	**0.7262**
Hubei	STARFM	0.8252	0.0400	0.0334	0.7529
	FSDAF	0.8214	0.0402	0.0333	0.7522
	CAFE	0.8498	0.0295	0.0213	0.7915
	ECPW-STFN	0.8475	0.0287	0.0213	0.7786
	DCAF	**0.8550**	**0.0284**	**0.0202**	**0.7924**

learning techniques to preserve spectral fidelity. The proposed DCAF method achieved the best overall performance. Specifically, for the BZH dataset, DCAF outperformed all competing methods in terms of the RMSE, MAE, and SSIM, surpassing the best three-input method (CAFE) by 6.8%, 8.7%, and 1.0%, respectively. Although its CC was slightly lower than that of CAFE, DCAF still improved CC by 1.1% compared with the other two-input method, ECPW-STFN.

The HRB region, characterized by arid surface conditions and relatively simple land features, presents a more challenging environment for spatiotemporal fusion. While the overall performance of all methods declined in this region, our method maintained a clear advantage. Experimental results

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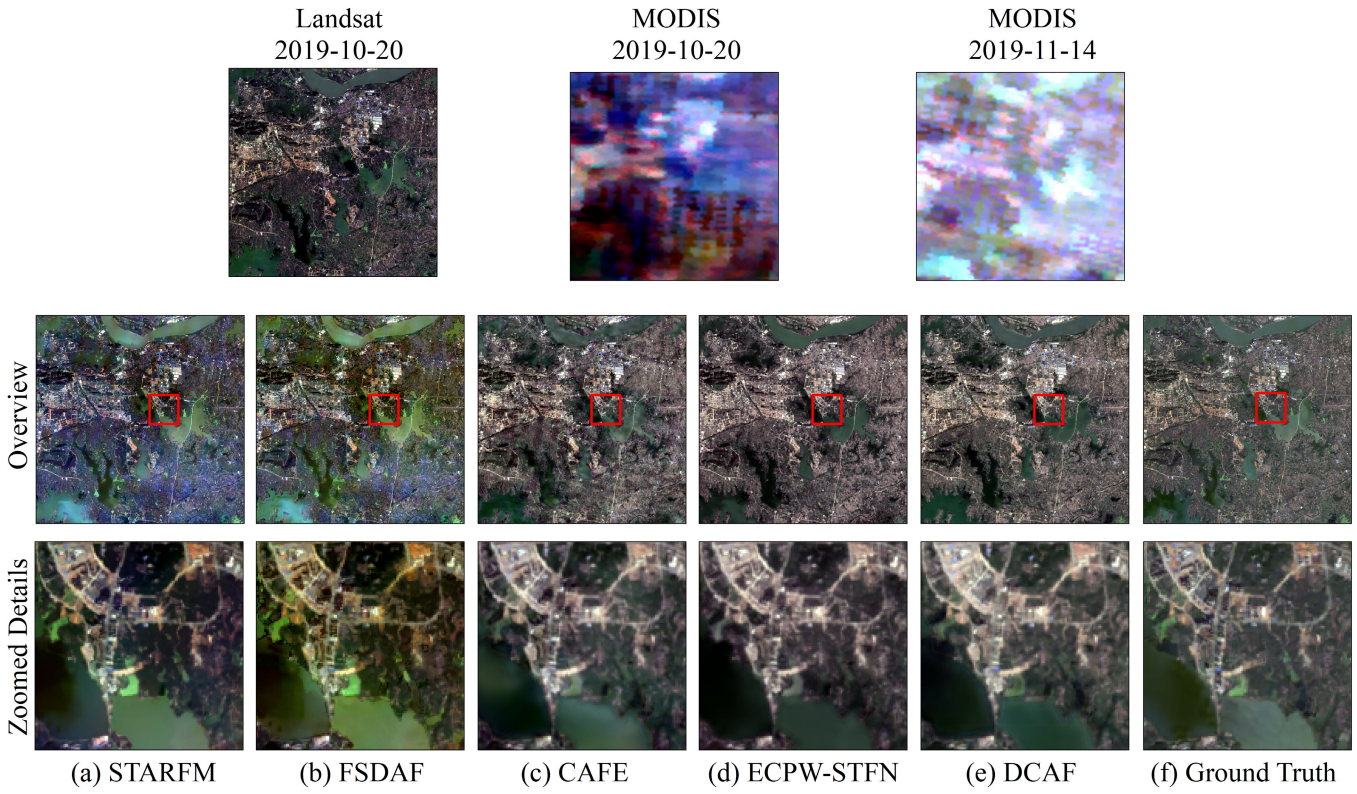


Fig. 8. Fusion results for the Hubei area on November 14, 2019, obtained using different spatiotemporal fusion methods. The first row shows the input data, the second row presents the RGB composites of the fusion results, and the third row displays enlarged subregions highlighted by red boxes in the second row. (a) STARFM. (b) FSDAF. (c) CAFF. (d) ECPW-STFN. (e) DCAF. (f) Ground truth.

indicate that FSDAF performed slightly worse than STARFM, primarily because it is more sensitive to sparse vegetation, which obscured class distinctions and degraded fusion accuracy. Compared with the best-performing three-input method (CAFE), DCAF improved the CC, RMSE, MAE, and SSIM by 3.3%, 3.6%, 5.3%, and 1.3%, respectively. ECPW-STFN performed slightly below CAFE, primarily due to ambiguous land-cover features that reduced its predictive precision. In general, the sparse and weakly dynamic vegetation in the HRB area increased the difficulty of prediction, as both CAFE and ECPW-STFN struggled with high spatial heterogeneity despite partially capturing temporal variations. By contrast, the proposed DCAF method demonstrates strong adaptability to vegetation change and spatial complexity, achieving the best performance across all evaluation metrics. These results further validate the robustness and generalization capability of the proposed method across different environmental conditions.

Hubei dataset presents substantial challenges for spatiotemporal image fusion due to its diverse land-cover types and complex spatial textures. As shown in Table II, the proposed DCAF method demonstrates superior performance in this region, achieving the highest accuracy and lowest overall error among all methods. It consistently outperformed all benchmark models, regardless of whether they utilize two or three input images. Compared with the CAFE method, DCAF improved CC by 0.6%, reduced RMSE and MAE by 1.0% and 5.2%, respectively, and increased SSIM by 0.1%. These experimental results indicate that the proposed model outperforms baseline methods across multiple evaluation metrics, demonstrating superior capability in both fine-detail reconstruction and global consistency preservation.

C. Application Evaluation

To assess the potential applicability of the fusion results for regional land-cover characterization, the normalized difference vegetation index (NDVI) and normalized difference water index (NDWI) were analyzed for the BZH and Hubei regions. These indices were not evaluated for the HRB region due to the absence of distinct vegetation and water features.

As illustrated in Fig. 9, all methods produced generally satisfactory NDVI and NDWI fusion results in the BZH area, with spatial patterns highly consistent with those derived from Landsat images. For NDVI mapping, STARFM and FSDAF tended to underestimate vegetation values, particularly along water body boundaries, whereas ECPW-STFN exhibited spatially inconsistent errors with relatively large fluctuations across regions. In contrast, both CAFE and the proposed DCAF method achieved more accurate NDVI estimates across most areas, although relatively large underestimation persisted in water bodies. Regarding NDWI, the baseline methods showed varying degrees of discontinuity in small tributary detection, with ECPW-STFN exhibiting the most severe fragmentation. By comparison, the proposed DCAF method achieved superior accuracy in water body delineation,

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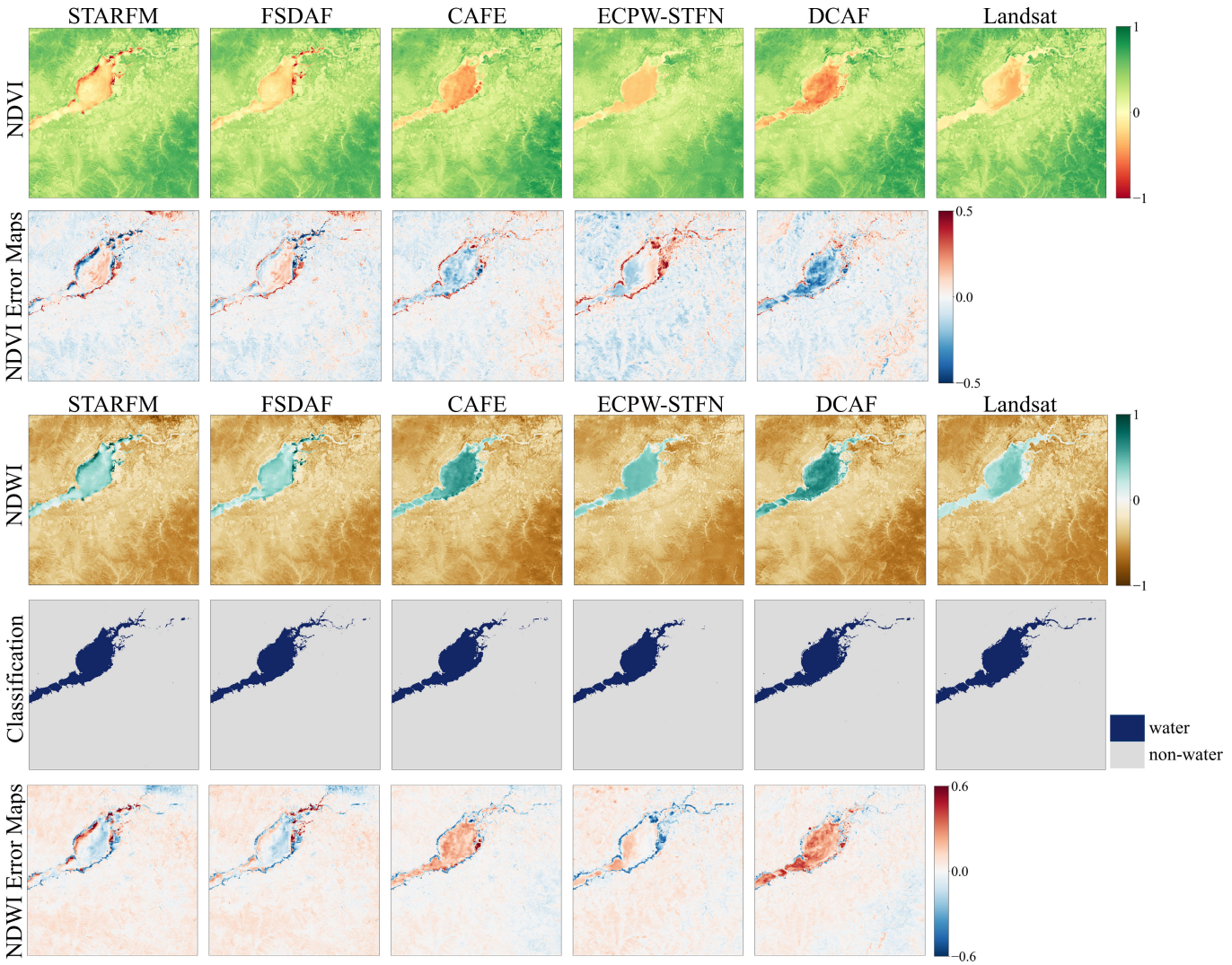


Fig. 9. NDVI and NDWI results derived from different fusion methods for the BZH area on April 29, 2020. The first row shows the NDVI maps, and the second row presents the differences between the NDVI values from each fusion model and the Landsat reference. The third row shows the NDWI maps, the fourth row presents the binary classification results for water bodies, and the fifth row illustrates the differences in NDWI values between each fusion model and the Landsat reference.

producing spatially continuous and coherent NDWI patterns. However, both CAFE and DCAF exhibited relatively large overestimation of NDWI values within water bodies.

Fig. 10 shows a comparative analysis of NDVI and NDWI results generated by different methods for the Hubei area on November 14, 2019. For NDVI, STARFM and FSDAF exhibited pronounced underestimation, while the three deep-learning-based methods achieved higher accuracy. For NDWI, STARFM and FSDAF showed notable misclassification, incorrectly identifying portions of land as water bodies, while ECPW-STFN performed poorly in extracting small water bodies. In contrast, the proposed DCAF method exhibited clear advantages in preserving spatial detail and delineating edges, accurately capturing the boundary characteristics and spatial distribution of water bodies. These improvements are particularly evident in complex surface regions characterized by mixed vegetation and water cover.

We conducted quantitative evaluations to further demonstrate the applicability of the fusion results for NDVI and NDWI mapping (Fig. 11). Given that this section focuses on vegetation areas, the evaluation range of NDVI is limited to $[0, 1]$ to emphasize effective vegetation signals, whereas NDWI is analyzed over its full range of $[-1, 1]$ to comprehensively assess the model's ability to distinguish between water and nonwater areas. For NDVI in the BZH region, although the ECPW-STFN method exhibited a concentrated distribution between reconstructed and actual values, its low R^2 and high RMSE indicate limited statistical accuracy. In contrast, the CAFE method achieved the best performance in both R^2 and RMSE. Notably, while the proposed DCAF method performed slightly below the three-input CAFE method, it still surpassed the two-input ECPW-STFN method and the traditional fusion methods (STARFM and FSDAF). In the NDWI analysis of the BZH region, DCAF exhibited a distinct advantage, with R^2 values improving by 0.42%–6.00% and the RMSE decreasing

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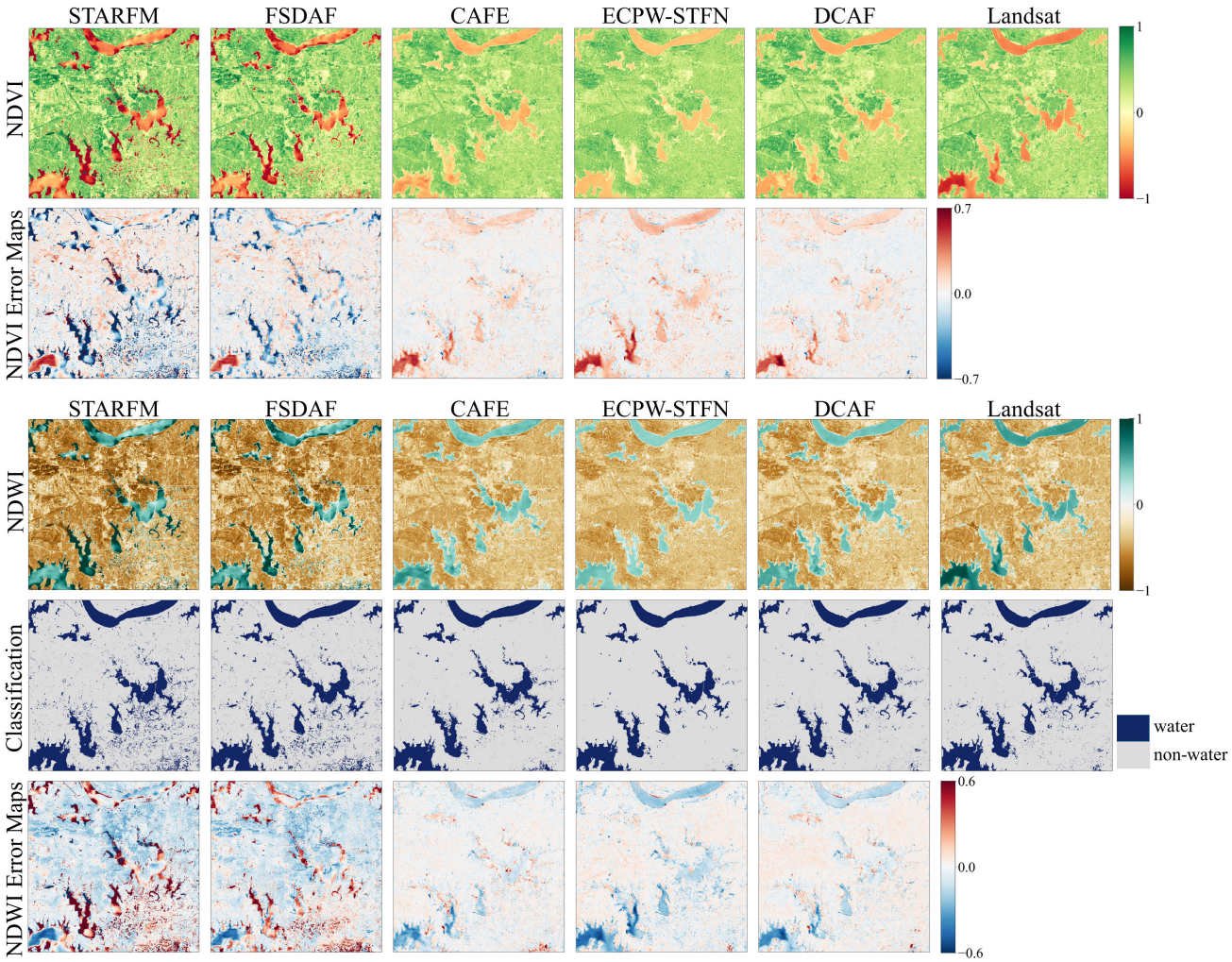


Fig. 10. NDVI and NDWI results derived from different fusion methods for the Hubei area on November 14, 2019. The layout of the results follows the same format as Fig. 9.

by 2.48%–22.73% compared to other methods, indicating substantially higher accuracy.

In the Hubei region, the CAFE method achieved the best NDVI performance, followed closely by the proposed DCAF method. Both models demonstrate strong potential for accurate NDVI extraction. Compared with the two-input ECPW-STFN method and the traditional fusion methods (STARFM and FSDAF), DCAF achieved significant improvements in both R^2 and RMSE metrics—specifically, increasing R^2 by 5.94% and reducing RMSE by 3.05% relative to ECPW-STFN. In the NDWI analysis, DCAF further underscores its superiority, achieving the highest R^2 and the lowest RMSE among all methods. Compared with the four baseline methods, DCAF improved R^2 by 0.09%–16.17% and reduced RMSE by 0.52%–34.17%, confirming its effectiveness and robustness in extracting remote sensing indices. Overall, the proposed DCAF method demonstrates significant advantages in both the BZH region, characterized by limited water coverage, and the Hubei region, featuring abundant water bodies. These results highlight its robustness and adaptability across diverse geographic environments and surface conditions, making it suitable for high-quality data fusion in complex scenarios.

Notably, DCAF adopts a two-input framework, utilizing a MODIS target image and a Landsat reference image, whereas CAFE incorporates an additional MODIS reference image. The extra observation in CAFE provides richer temporal constraints, which may slightly improve certain metrics. However, despite relying on fewer inputs, DCAF achieves comparable or superior performance on most quantitative metrics due to its improved cross-attention fusion framework, residual dense attention mechanism, and adaptive temporal-difference weighting strategy. Furthermore, NDWI results demonstrate that DCAF more effectively preserves water-related spectral features. These findings highlight that DCAF can efficiently exploit the complementary strengths of MODIS and Landsat while maintaining high reconstruction accuracy with reduced data dependency.

V. Discussion

A. Ablation Experiments

To assess the contributions of the MDCA, GFCA, RDAM, and DWFM modules to overall model performance, four ablation experiments were conducted. Each module was systematically removed, and the resulting fusion performance was

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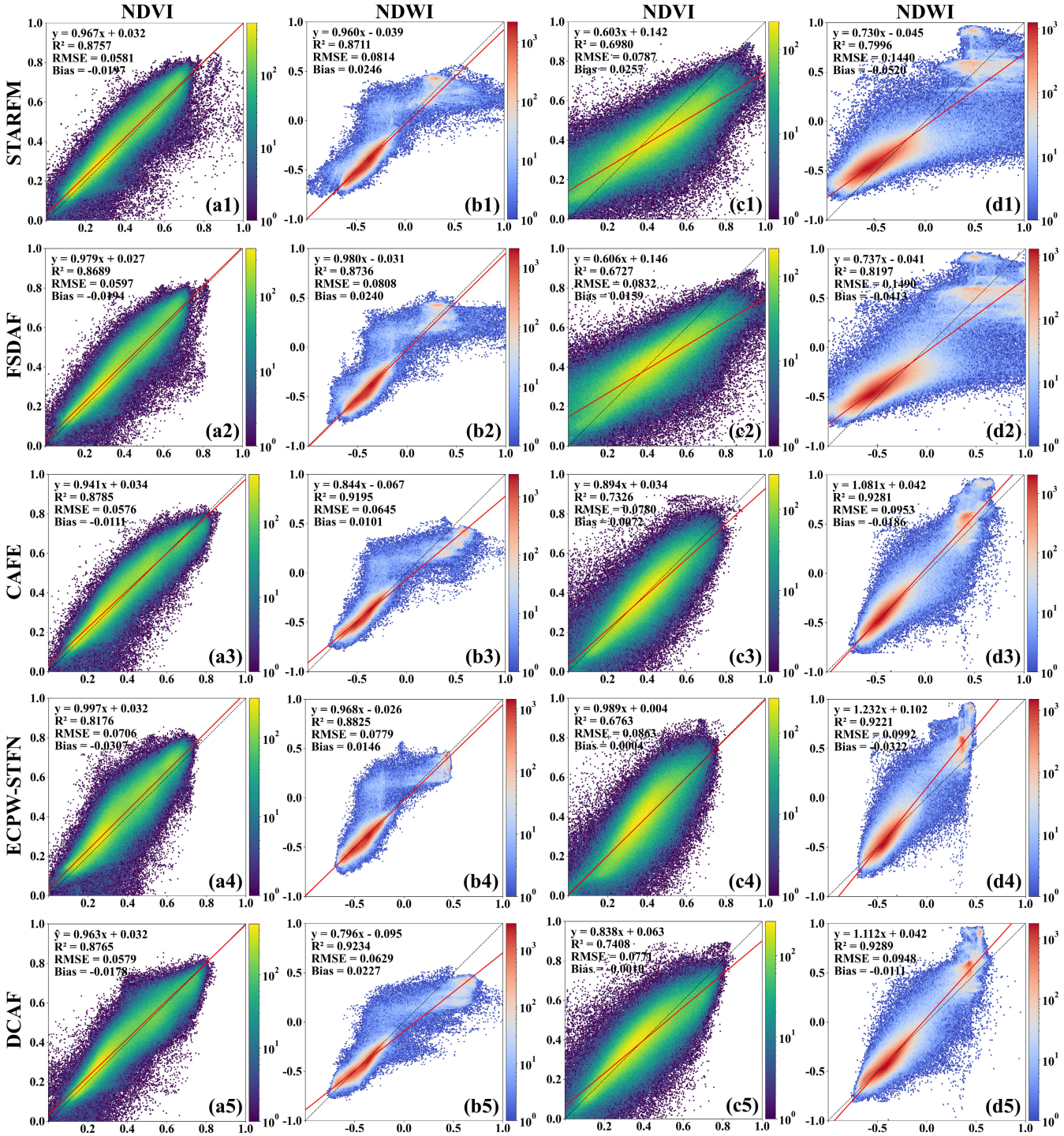


Fig. 11. Scatter density plots of NDVI and NDWI results obtained using different fusion methods in the BZH and Hubei regions: (a1)–(a5) NDVI for the BZH area, (b1)–(b5) NDWI for the BZH area, (c1)–(c5) NDVI for the Hubei area, and (d1)–(d5) NDWI for the Hubei area.

quantitatively evaluated across the BZH, HRB, and Hubei regions, as summarized in Table III. The results indicate that the removal of any module led to a decrease in model performance, confirming the effectiveness of each component in enhancing fusion accuracy.

Among all the modules, GFCA had the greatest impact on model performance in both the BZH and Hubei regions. Its exclusion increased the RMSE and MAE by approximately 13.67% and 16.02% in the BZH area and by 14.79% and 19.31% in the Hubei area, respectively. GFCA focuses on capturing intrinsic interband dependencies and enhancing global spectral consistency, which is particularly advantageous in regions with diverse land-cover compositions.

Removing the MDCA module causes the most pronounced performance degradation in the HRB region compared with the other study areas, particularly in terms of CC. This suggests

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TABLE III
QUANTITATIVE EVALUATION RESULTS OF THE ABLATION EXPERIMENTS FOR THE THREE STUDY AREAS

Area	Indicator	DCAF	MDCA-removed	GFCA-removed	RDAM-removed	DWFM-removed
BZH	CC	**0.8946**	0.8915	0.8806	0.8902	0.8911
	RMSE	**0.0300**	0.0317	0.0341	0.0312	0.0326
	MAE	**0.0231**	0.0247	0.0268	0.0242	0.0255
	SSIM	**0.8138**	0.8040	0.7998	0.8049	0.8090
HRB	CC	**0.7400**	0.7255	0.7280	0.7330	0.7360
	RMSE	**0.0317**	0.0334	0.0322	0.0329	0.0326
	MAE	**0.0234**	0.0245	0.0238	0.0244	0.0242
	SSIM	**0.7262**	0.7212	0.7251	0.7250	0.7211
Hubei	CC	**0.8550**	0.8507	0.8478	0.8493	0.8517
	RMSE	**0.0284**	0.0295	0.0326	0.0288	0.0317
	MAE	**0.0202**	0.0212	0.0241	0.0205	0.0236
	SSIM	**0.7924**	0.7901	0.7850	0.7907	0.7870

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF THE LOSS FUNCTION ABLATION EXPERIMENTS FOR THE THREE STUDY AREAS

Area	Indicator	DCAF	Charbonnier-removed	Edge-removed	Structure-removed
BZH	CC	**0.8946**	0.7242	0.8698	0.8916
	RMSE	**0.0300**	0.3745	0.0356	0.0313
	MAE	**0.0231**	0.3673	0.0278	0.0244
	SSIM	**0.8138**	0.6479	0.7982	0.8090
HRB	CC	**0.7400**	0.6051	0.7256	0.7317
	RMSE	**0.0317**	0.4736	0.0324	0.0321
	MAE	**0.0234**	0.4666	0.0239	0.0236
	SSIM	**0.7262**	0.5815	0.7207	0.7245
Hubei	CC	**0.8550**	0.7098	0.8541	0.8497
	RMSE	**0.0284**	0.3179	0.0323	0.0322
	MAE	**0.0202**	0.3099	0.0241	0.0238
	SSIM	**0.7924**	0.6350	0.7863	0.7850

that MDCA plays a more critical role in maintaining feature consistency under complex and heterogeneous conditions. Although its contribution is relatively moderate in terms of the RMSE and MAE, its consistent impact across all metrics indicates that MDCA primarily enhances adaptive feature refinement rather than directly reducing reconstruction error. In contrast, the GFCA module exhibits a comparable yet more stable contribution, highlighting its role in capturing global spectral information and ensuring overall consistency. These observations suggest that MDCA and GFCA serve complementary functions: MDCA focuses on adaptive recalibration of local features, while GFCA facilitates global feature aggregation. Their integration enables the model to jointly preserve global spectral consistency and local structural fidelity, thereby improving robustness across diverse surface conditions.

The RDAM module introduces lightweight feature refinement to enhance model stability and improve generalization across different regions, resulting in consistently minor variations in performance across all evaluation metrics. In contrast,

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the DWFM module performs adaptive difference-weighted feature fusion to enhance local temporal consistency. Its removal leads to more noticeable degradation in error-based metrics (e.g., RMSE and MAE), particularly in complex regions such as BZH and Hubei, indicating its effectiveness in improving quantitative reconstruction accuracy. However, its influence on SSIM metrics remains relatively limited.

In summary, the integration of all four modules within the proposed DCAF framework consistently improves fusion accuracy and robustness across diverse environmental conditions.

B. Effectiveness of Loss Functions

The effectiveness of the loss functions was assessed, and the results are summarized in Table IV. The Charbonnier loss function plays a pivotal role in the DCAF model, substantially enhancing robustness and noise resistance, particularly when handling complex data distributions in the BZH, HRB, and Hubei regions. The Edge loss function has the most significant impact on the BZH region—its removal resulted in decreases of approximately 2.77% in the CC and 1.92% in the SSIM, along with increases of 18.67% and 20.35% in the RMSE and MAE, respectively. These findings emphasize the critical role of the Edge loss function in preserving edge information and improving fusion accuracy. In comparison, the Structure loss function has a relatively minor impact on overall performance but still contributes to maintaining structural similarity and spatial coherence. Overall, the combined use of the Charbonnier, Edge, and Structure loss functions enables the DCAF model to achieve superior predictive performance and strong generalization across all study regions. These results underscore the importance of the integrated loss design in enhancing both the accuracy and robustness of the proposed model.

VI. Conclusion

This study proposes a deep fusion network, DCAF, which can generate high-resolution target images using only cross-pair inputs from Landsat and MODIS data. The DCAF framework integrates a multiscale attention mechanism, multilevel feature enhancement, and adaptive difference weighting fusion. Specifically, the MDCA and GFCA modules effectively leverage the complementary multiscale information from low- and high-resolution data through multitype cross-attention mechanisms. The RDAM module further captures hierarchical feature relationships, enhancing overall feature representation, while the DWFM module facilitates effective fusion by incorporating a temporal difference weighting strategy that effectively captures local change information.

The proposed DCAF method is comprehensively evaluated through both qualitative and quantitative comparisons with four baseline models. While DCAF is slightly outperformed by the three-input CAFE model in specific metrics within the BZH region, it achieves superior fusion accuracy and application performance in the other areas. Moreover, DCAF consistently outperforms the other two-input methods across all evaluated regions. Overall, the proposed fusion method exhibits strong potential for generating high-resolution data across regions with diverse surface and environmental conditions. Nevertheless, the fusion accuracy of DCAF remains constrained in areas characterized by abrupt surface changes and high spatial heterogeneity. Future work will focus on enhancing the model's adaptability to such complex environments.

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